

TE Curve Verification Pipeline Original ONNX Forward Collage Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates only the recovered paper-original forward ONNX model bank for paper_original_best_Fw . The curve prediction is rebuilt directly from the 19 original target models under reference/rcim_ml_compensation_recovered_assets/models/exact_onnx_paper_release .

Loaded ONNX Targets

The table lists the original release filenames loaded for the 19 paper-best forward targets. Full original ONNX paths are kept in the validation summary YAML.

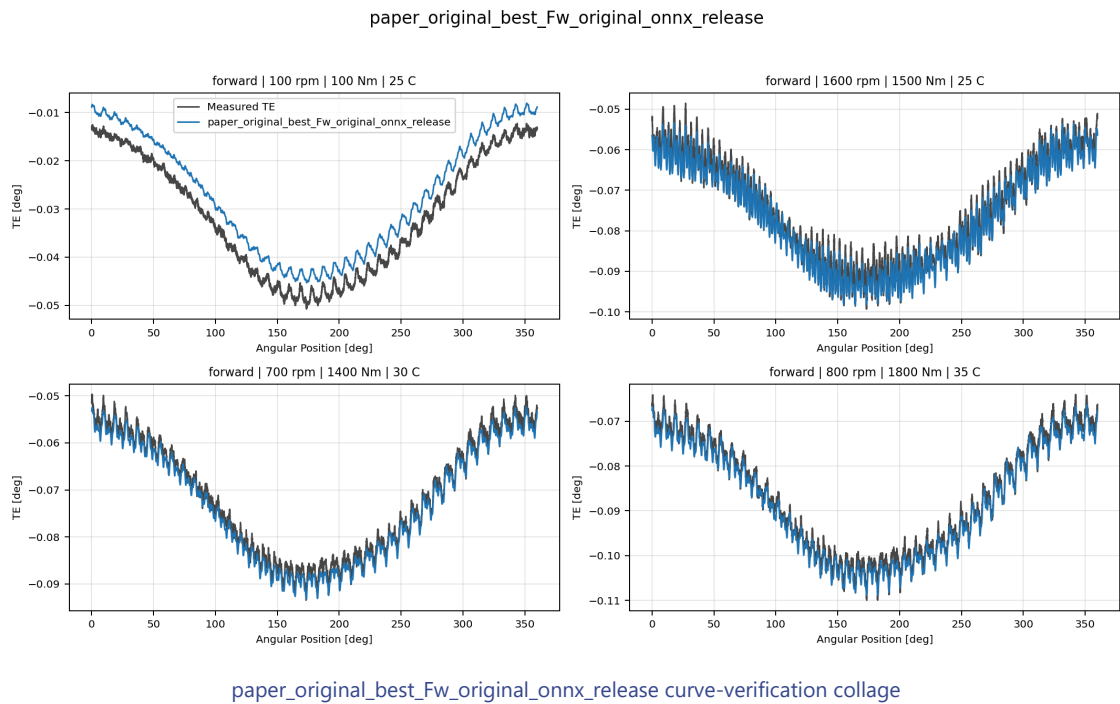
Target	Harmonic	Family	Original ONNX Files
amplitude	A0	SVR	SVR_ampl0.onnx
amplitude	A1	RF	RandomForestRegressor_ampl1.onnx
amplitude	A3	HGBM	HistGradientBoostingRegressor_ampl3.onnx
amplitude	A39	HGBM	HistGradientBoostingRegressor_ampl39.onnx
amplitude	A40	ERT	ExtraTreesRegressor_ampl40.onnx
amplitude	A78	HGBM	HistGradientBoostingRegressor_ampl78.onnx
amplitude	A81	RF	RandomForestRegressor_ampl81.onnx
amplitude	A156	ERT	ExtraTreesRegressor_ampl156.onnx
amplitude	A162	ERT	ExtraTreesRegressor_ampl162.onnx
amplitude	A240	ERT	ExtraTreesRegressor_ampl240.onnx
phase	P1	LGBM	LGBMRegressor_phase1.onnx
phase	P3	HGBM	HistGradientBoostingRegressor_phase3.onnx
phase	P39	HGBM	HistGradientBoostingRegressor_phase39.onnx
phase	P40	GBM	GradientBoostingRegressor_phase40.onnx
phase	P78	RF	RandomForestRegressor_phase78.onnx
phase	P81	RF	RandomForestRegressor_phase81.onnx
phase	P156	RF	RandomForestRegressor_phase156.onnx
phase	P162	ERT	ExtraTreesRegressor_phase162.onnx
phase	P240	ERT	ExtraTreesRegressor_phase240.onnx

TE Curve Verification Pipeline Forward Metrics

Candidate	Curves	MAE [deg]	RMSE [deg]	Mean Error [%]	P95 Error [%]
paper_original_best_Fw ONNX	97	0.002804	0.002987	6.329	13.847

Collage

The collage shows four deterministic held-out forward `TE Curve Verification Pipeline` curves spread across the sorted forward evaluation set.



Collaged Curves

Curve	Speed [rpm]	Torque [Nm]	Oil [C]	MAE [deg]	Mean Error [%]
Curve 1	100	100	25	0.004471	11.607
Curve 2	1600	1500	25	0.001935	3.809
Curve 3	700	1400	30	0.001359	3.104
Curve 4	800	1800	35	0.001279	2.786

Output Artifacts

- output directory: `output\validation_checks\track2_original_onnx_fw_collage_report\2026-06-08-12-57-36__track2_original_onnx_fw_collage_report` ;
- summary YAML: `output\validation_checks\track2_original_onnx_fw_collage_report\2026-06-08-12-57-36__track2_original_onnx_fw_collage_report\track2_original_onnx_fw_collage_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_original_onnx_fw_collage_report\2026-06-08-12-57-36__track2_original_onnx_fw_collage_report\track2_original_onnx_fw_collage_metrics.csv` ;
- report Markdown: `doc\reports\analysis\track2\original_onnx_fw_collage_report\[2026-06-05]\track2_original_onnx_fw_collage_report.md` .